Hardware-Aware Quantum Bayesian Learner Compression: Linking Human Belief Updating to NISQ Circuit Complexity

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Abstract—This project connects quantum models of cognition with the engineering constraints of near-term quantum hardware. Bayesian theories treat human learning as belief updating over structured hypotheses, and quantum cognition represents belief states in Hilbert space with evidence acting through quantum channels. We instantiate a small quantum Bayesian learner: a belief state on a two-qubit register, a prior, and a stimulus-dependent evidence channel that updates the belief on a synthetic similarity-structured categorization task inspired by the simplicity principle of Pothos and Chater, with the posterior category read out by measurement. To tie this cognitive picture to real devices, we make the learner a hardware-efficient variational circuit and add a hardware-aware penalty equal to the number of two-qubit gates that survive transpilation to IBM's FakeManila coupling map, then compress the circuit by greedy structured pruning of individual Heisenberg interaction terms. The central question is where the capacity boundary lies: how much entangling structure can be removed before the learner can no longer accurately represent the task. Across three task difficulty levels and five seeds we trace accuracy–complexity frontiers and find that structured compression is essentially free, and on the hardest task beneficial (accuracy peaks at a quarter of the full two-qubit cost), while removing all entanglers collapses the learner to chance—a concrete cognitive-capacity boundary. Learned masks beat random masks at every matched budget (by roughly 6–12 percentage points on the hard task), so which interactions are kept matters, not only how many. Finally, on a physical IBM device (ibm_fez) the full circuit collapses to near chance (0.60) while the compressed circuit retains accuracy (0.83): on real NISQ hardware, hardware-aware compression of a quantum Bayesian learner is not merely economical but necessary.

Index Terms—Quantum cognition, Bayesian learning, belief updating, quantum channels, variational quantum circuits, hardware-efficient ansatz, NISQ devices, circuit compression.

I. Introduction

Bayesian models are among the most influential accounts of human learning: a learner holds a prior over hypotheses, receives evidence, updates its beliefs, and gradually converges on a better model of the world. This view has been developed in detail for concept learning and cognitive development [1], [2], [11], where structured priors and likelihoods explain how people generalize from few examples. A younger but active direction, quantum cognition, represents belief states in Hilbert space—as density operators—and models judgment and decision making with the mathematics of quantum theory, capturing interference and order effects that are awkward for classical probability [3]. Related work shows how probabilistic inference and Bayesian networks can be embedded directly into quantum circuits, encoding distributions and conditional dependencies into states that are manipulated by unitary or measurement-based operations [8], [9].

In parallel, noisy intermediate-scale quantum (NISQ) devices [7] have made variational quantum algorithms (VQAs) the dominant near-term paradigm [6], in which a parameterized circuit is trained by a classical optimizer. The choice of ansatz is central: hardware-efficient ansätze, built from single-qubit rotations and native two-qubit entanglers matched to a device coupling graph [4], trade analytic structure for shallow depth, and their expressivity and entangling capability have been studied carefully [5], [27]. Two-qubit gates dominate the error budget on current hardware [7], [10], so reducing their number without destroying task performance is a central engineering objective—one that quantum advantage arguments make only more pressing as circuits grow [20], [21].

This work joins the two threads. We build a small quantum Bayesian learner whose belief state is a density operator on a two-qubit register and whose incorporation of evidence is a stimulus-dependent quantum channel [3], [12], [13], trained on a toy categorization task in the spirit of similarity-based human categorization [19]. We then subject the learner to realistic hardware pressure: the update circuit is a hardware-efficient, Heisenberg-type ansatz, and we add a hardware-aware loss that penalizes the number of two-qubit gates remaining after transpilation to a target backend. By training and compressing this learner we obtain an empirical trade-off between task performance—accuracy and cross-entropy on the categorization task—and circuit capacity—the entangling structure of the underlying circuit under NISQ constraints—which we interpret through the lens of cognitive capacity: the learner's ability to update beliefs and categorize stimuli. The guiding question is concrete: how much of the circuit can we remove before the quantum Bayesian learner can no longer represent the categorization task?

Our contributions are: (1) a compact quantum Bayesian learner, framed in the density-operator / quantum-channel language of quantum cognition and realized as an exactly-simulated, analytically-differentiable circuit that successfully learns the task; (2) a hardware-aware cost based on the real post-transpile two-qubit count on FakeManila, resolved at the level of individual $R_{XX}/R_{YY}/R_{ZZ}$ interaction terms; (3) empirical accuracy–complexity frontiers, across task difficulty levels, that locate a cognitive-capacity boundary and show that structured compression is free or beneficial; (4) a learned-versus-random ablation establishing that the pattern of retained entanglers drives performance; and (5) a validation on a physical IBM device showing that on real hardware compression is what makes the learner usable at all.

II. Methods and Model

A. Quantum Bayesian Belief States and Evidence Channels

Following the density-operator framework of quantum cognition [3] and the theory of quantum channels [12], [13], [26], the learner's belief about a stimulus is a state ρ on an n-qubit register. The prior is a fixed "no evidence" state ρ_0, and the incorporation of a stimulus x is modeled as a completely-positive trace-preserving (CPTP) map—a quantum channel—conditioned on x:

$$\rho \;\longrightarrow\; \mathcal{E}_x(\rho) = \sum_k E_{x,k}\, \rho\, E_{x,k}^\dagger, \qquad \sum_k E_{x,k}^\dagger E_{x,k} = \mathbb{I}, \tag{1}$$

where $\{E_{x,k}\}$ are stimulus-dependent Kraus operators [12], [13]. This mirrors how quantum-cognitive models implement belief updates through channels and measurement-like operations [3], [8], [9]. The posterior category probability is then obtained by measuring the updated belief state (Section II-D). We use "Bayesian" here in the operational sense of belief-state updating under evidence—a prior state revised by a stimulus-conditioned evidence channel—rather than as an explicit symbolic inference over an enumerated hypothesis set with likelihoods.

We provide two realizations of this learner. A fast pure-state realization takes $\rho_0 = |0\rangle\langle 0|^{\otimes n}$ and a unitary evidence channel $\mathcal{E}_x(\rho) = U(x;\theta,m)\,\rho\, U(x;\theta,m)^\dagger$ built from a data re-uploading encoding interleaved with the masked Heisenberg layers of Section II-B; the belief stays pure, so it is exactly simulable as a statevector and is used for the bulk of the experiments below. A second, mixed-state realization implements the channel (1) literally: starting from the maximally mixed prior $\rho_0 = \mathbb{I}/2^n$ (a uniform "no information" belief), each update applies the unitary layers and then a genuinely non-unital evidence channel—a stimulus-dependent Lüders/POVM filter $K_x \rho K_x^\dagger$ followed by the Bayesian trace renormalization $\rho \mapsto \rho/\mathrm{Tr}[K_x \rho K_x^\dagger]$, together with per-qubit amplitude damping (a multi-Kraus, non-unital CPTP map). Here the belief is a genuine mixed density operator (purity < 1, verified positive semidefinite and trace-one); crucially, because a unitary leaves $\mathbb{I}/2^n$ invariant, it is precisely the non-unitality of the evidence channel that lets evidence move the belief, mirroring the role of evidence in Bayesian updating. Both realizations are exactly simulated and analytically differentiable, and the mixed-state model reduces bit-for-bit to the pure-state one when the channel is switched off; Section IV-F shows it reproduces the accuracy–complexity findings, so the conclusions do not depend on the pure-state simplification.

B. Variational Ansatz and Heisenberg Entanglers

The update map $U(x;\theta,m)$ is a layered hardware-efficient circuit consistent with ansätze used in VQE and related methods [4]–[6], [25]. Data re-uploading re-encodes the stimulus across the depth, giving the small circuit nontrivial expressive power [5], [27]. Each of the D layers applies single-qubit R_X, R_Z rotations on every qubit, followed by a Heisenberg-type entangling block on each connected pair (i,j),

$$U_{ij}(\alpha,\beta,\gamma) = \exp[-i\,(\alpha X_i X_j + \beta Y_i Y_j + \gamma Z_i Z_j)]\,, \tag{2}$$

realized as native R_{XX}, R_{YY}, R_{ZZ} gates; such two-body spin interactions are standard in VQE simulations of magnets [14], [15]. The main experiments use $n = 2$ qubits and depth $D = 4$ with two re-uploads.

C. Per-Term Masking and Hardware-Aware Cost

The entangling structure is learnable. A binary mask $m_{\ell,p} \in \{0,1\}$ selects each interaction term $p \in \{XX, YY, ZZ\}$ in each layer ℓ ($R_{PP}(0) = \mathbb{I}$ when off), giving 12 maskable terms for $D = 4$ and one pair—a small, task-specific instance of quantum architecture search [16], [17]. The hardware cost of a masked circuit is the number of two-qubit gates remaining after transpilation to IBM's 5-qubit FakeManila backend (including SWAPs inserted for the coupling map), computed by actually invoking Qiskit's transpiler [24]; it is deterministic and resolved per term (under the transpiler basis and optimization settings used here, each active Heisenberg term costs two CNOTs, so the full circuit gives $N_{2q} = 24$). The learning objective is

$$L(\theta,m) = \frac{1}{N}\sum_{k=1}^{N} \ell\big(y_k, \hat{y}(x_k;\theta,m)\big) + \lambda\, N_{2q}(m), \tag{3}$$

with ℓ the binary cross-entropy and N_{2q} the real post-transpile count, motivated by the dominance of two-qubit error on NISQ devices [7], [10]. The mask sparsity is $s =$ (active terms)/12.

D. Measurement Readout

The category-1 probability is read from single-qubit Pauli expectations of the updated belief, $v = (\langle Z_0\rangle, \langle Z_1\rangle, \langle X_0\rangle, \langle X_1\rangle)$, through a trainable head $\hat{y} = \sigma(w \cdot v + b)$. Restricting the readout to single-qubit observables is deliberate: a product (unentangled) belief factorizes these expectations, so any feature interaction in the decision boundary must be supplied by the evidence channel's entangling layers. Entanglement is thus strictly necessary for nontrivial categorization, and the two-qubit budget meaningfully bounds the learner's cognitive capacity.

E. Optimization and Greedy Structured Compression

The continuous parameters θ (rotations and readout head) are trained by Adam with exact gradients obtained in JAX by differentiating directly through the exact belief-state simulator (no parameter-shift rule or finite differences). Compression uses greedy structured pruning: from the fully entangled circuit, we repeatedly remove the interaction term whose removal least increases training cross-entropy, retraining θ after each removal. This traces a trajectory of circuits from $N_{2q} = 24$ down to 0—the accuracy–complexity frontier—and the hardware-aware objective (3) selects an operating point along it for each λ. The greedy mechanism is stable and aligned with structured architecture search [16], [17] and pruning-based circuit optimization [29]; the shallow, small-register design keeps optimization well-conditioned against barren plateaus [18]. Because the loss landscape is non-convex, different mask configurations can in principle converge to different local optima; averaging over five random seeds—each greedy step warm-started from the previous solution—mitigates, but does not eliminate, this source of variability, which the reported standard deviations make explicit.

F. Controllable Categorization Task

To expose rather than saturate the capacity boundary we use a synthetic, similarity-structured checkerboard categorization task inspired by the simplicity principle that Pothos and Chater study for unsupervised human categorization [19] (it is a controlled synthetic task, not their behavioral dataset): stimuli are points in $[0, 1]^2$ and the category is the parity of a checkerboard cell, $y = (\lfloor fx_0\rfloor + \lfloor fx_1\rfloor) \bmod 2$. Nearby stimuli usually share a category (local similarity), but membership depends on the joint configuration of both features, so the boundary requires the feature interaction only entanglement can provide. The frequency f is a difficulty knob: $f = 2$ (easy), $f = 3$ (medium), $f = 4$ (hard) demand progressively more entangling capacity. Each dataset has 200 stimuli, split 70/30 with a fixed seed; classes are balanced, so chance is 0.5.

III. Experimental Setup

The learner is implemented as an exact, differentiable belief-state simulator; two-qubit cost is the real transpiled count on FakeManila [24]. Unless noted, the configuration is $n = 2$, $D = 4$, one qubit pair, two re-uploads, single-qubit-Pauli readout, and Heisenberg entanglers (12 maskable terms, $N_{2q} \in \{0, 2, \ldots, 24\}$). For each difficulty we run five seeds; each produces a greedy frontier, a λ-selected operating point for $\lambda \in \{0, 0.005, 0.01, 0.02, 0.04, 0.08, 0.15\}$, and, at every interior budget, three random masks of matched sparsity as a connectivity-matched control. All accuracy, cross-entropy, and N_{2q} values use the discrete deployed circuit and the real transpiled count, averaged over the five seeds. The software stack is Python 3.14, Qiskit 2.4.2, Qiskit Aer 0.17.2, Qiskit IBM Runtime 0.47.0, JAX 0.10.2, and scikit-learn 1.9.0 (exact pins in requirements.txt). As a difficulty sanity check, standard classical classifiers on the identical splits confirm the frequency knob is a real difficulty axis (Appendix B): a linear model stays at chance on every level, while kernel-SVM and MLP accuracy fall monotonically from easy to hard. Code and logs are in the public repository.[1]

IV. Results and Data Analysis

A. The Capacity Boundary: Accuracy–Complexity Frontier

Figure 1 plots mean test accuracy against the post-transpile two-qubit count N_{2q} for the three difficulties. The learner categorizes well above the 0.5 chance line at all nonzero budgets (easy/medium near 0.95–0.97, hard near 0.87–0.93); the frontiers are nearly flat over a wide range, so most entangling structure is redundant; and at $N_{2q} = 0$ every difficulty collapses toward chance (0.52 easy, 0.64 medium, 0.49 hard). This last point is the cognitive-capacity boundary made quantitative: an unentangled belief, read out by single-qubit measurements, cannot represent the categorization rule, so some entanglement is necessary to support belief updating, but far less than the full circuit provides.

[1] https://github.com/zoraizmohammad/qb-learner-compression

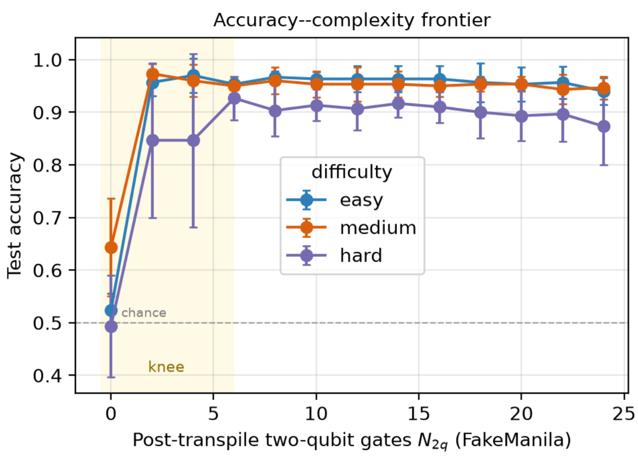

Fig. 1. Cognitive capacity versus circuit capacity. Test accuracy versus post-transpile two-qubit gate count N_{2q} on FakeManila (mean ± std, five seeds). All difficulties sit well above chance (dashed) for $N_{2q} > 0$ and collapse at $N_{2q} = 0$, while remaining nearly flat in between: the capacity boundary is sharp and lies at low N_{2q} (shaded knee), with the harder task requiring more entangling capacity.

The hard task is most informative because it is not saturated: accuracy peaks at an intermediate cost (0.927 at $N_{2q} = 6$) and is lower for the full circuit (0.873 at $N_{2q} = 24$). Moderate structured compression thus acts as a capacity-control regularizer, removing entangling capacity the optimizer would otherwise overfit, while the easier tasks have ample capacity and stay flat until the collapse at $N_{2q} = 0$.

B. Effect of the Hardware-Aware Penalty

Table I reports, for the hard task, the operating point selected by (3) as λ grows; the $\lambda = 0$ row is the no-penalty ideal baseline, the point chosen on accuracy alone. For small penalties the learner keeps a moderately rich circuit ($N_{2q} \approx 10$); as λ increases it prunes aggressively, reaching $N_{2q} = 3.2$ at $\lambda = 0.04$ with no loss of accuracy (a slight gain, $0.923 \to 0.933$), and only at the strongest penalty ($\lambda = 0.15$, $N_{2q} = 1.6$) does accuracy fall to 0.823. The learner sheds roughly two-thirds of its two-qubit gates for free and over 90% at modest cost—the penalty drives it toward the capacity boundary without crossing it.

TABLE I
Hardware-aware penalty sweep (hard task, five seeds). Operating point on the greedy frontier minimizing $L_{\mathrm{CE}} + \lambda N_{2q}$: mean two-qubit count, mask sparsity s, accuracy, cross-entropy.

λ	N_{2q}	s	Acc	CE
0.000	9.6	0.40	0.923	0.196
0.005	5.2	0.22	0.933	0.187
0.010	5.2	0.22	0.933	0.187
0.020	4.4	0.18	0.927	0.170
0.040	3.2	0.13	0.933	0.179
0.080	2.8	0.12	0.913	0.225
0.150	1.6	0.07	0.823	0.347

C. Structured vs Unstructured Capacity: Learned vs Random Masks

The frontier says how many entanglers are needed; the ablation in Fig. 2 and Table II asks whether which ones matter. At each budget we compare the learned (greedy) mask against random masks with the same number of active terms. On the hard task the learned mask wins at every budget (Table II), by roughly 6–12 percentage points, largest in the low-to-mid regime where structure is scarce and must be allocated well. Resolved per seed rather than aggregated, the learned mask beats the matched random masks in 49 of the 55 paired (seed, budget) comparisons, with a mean advantage of +0.083 and a one-sided Wilcoxon signed-rank $p < 10^{-7}$. In cognitive terms, capacity reductions that respect the task structure preserve belief updating, whereas unstructured loss of capacity removes the very interactions that carry the category boundary.

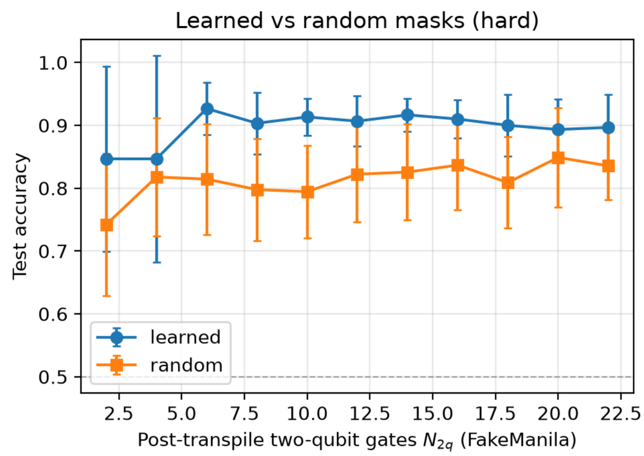

Fig. 2. Structured beats unstructured pruning (hard task, mean ± std, five seeds). At each matched two-qubit budget, greedy (learned) masks retain task-relevant interactions and outperform random masks of identical sparsity.

TABLE II
Learned vs random masks (hard task, mean accuracy, five seeds). Δ is the learned advantage at the same two-qubit budget.

N_{2q}	Learned	Random	Δ
2	0.847	0.742	+0.104
6	0.927	0.814	+0.112
10	0.913	0.794	+0.119
14	0.917	0.826	+0.091
18	0.900	0.809	+0.091
22	0.897	0.836	+0.061

D. Robustness Under a Device Noise Model

To check the frontier survives device imperfections, we re-evaluated the trained circuits—unchanged and bit-identical to the model—on Qiskit's FakeManila noise model at 4096 shots. Across five seeds (Fig. 3), accuracy under the noise model tracks the noiseless frontier within a few percent at every budget, with overlapping error bars: the cognitive-capacity trade-off persists under realistic noise, so the gate-count savings are genuine net of noise.

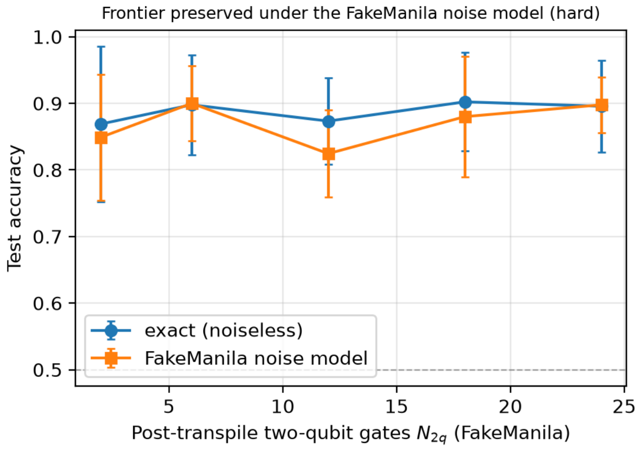

Fig. 3. Accuracy under the FakeManila noise model vs exact simulation (hard task, five seeds, 4096 shots). The curves overlap across all budgets.

E. Validation on a Physical Device

We then ran the trained circuits on a physical IBM Quantum processor (ibm_fez, a 156-qubit Heron device), mapping the accuracy–complexity frontier on-device across five two-qubit budgets at 2048–4096 shots (run provenance—date, shots, transpiler settings—in Appendix A). Each device point is a single trained model (seed 0) evaluated in one job on 16 held-out test stimuli at 2048 shots; the endpoint confirmation below uses 40 test stimuli at 4096 shots. Device cost limits the on-device sweep to five two-qubit budgets. The outcome (Fig. 4) is clear in this controlled run and reverses the simulated picture. In exact simulation the frontier is nearly flat; on hardware it slopes steeply downward as N_{2q} grows—device accuracy holds at 0.81 for the cheap circuits ($N_{2q} = 2, 6$), slips to 0.75 at $N_{2q} = 12$, and collapses to near chance (0.56) at $N_{2q} = 18, 24$, where the full circuit's two-qubit gates accumulate enough error to destroy the belief update. A separate denser run (40 test stimuli, 4096 shots) confirms the endpoints, with the full circuit at 0.60 (simulation 0.95) and the compressed circuit at 0.83 (simulation 0.93). On real NISQ hardware, therefore, hardware-aware compression of the quantum Bayesian learner is not merely free but necessary: more entangling capacity, helpful or neutral in simulation, is actively harmful on a device, and pruning it in a task-aware way is what makes the learner work at all.

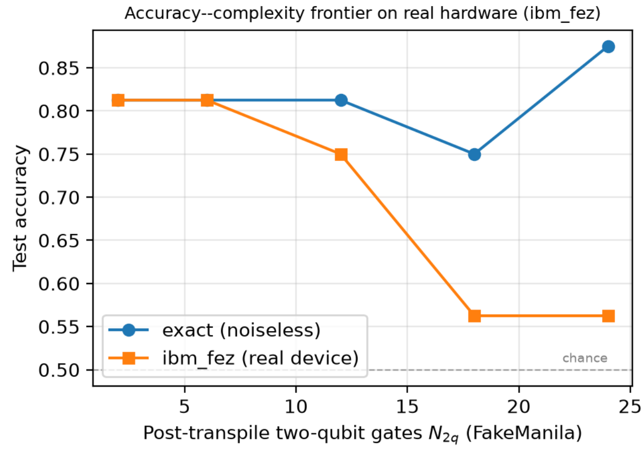

Fig. 4. The accuracy–complexity frontier on a physical IBM device (ibm_fez, hard task, 16 test stimuli, five device budgets). In exact simulation accuracy is flat across budgets; on hardware it falls monotonically with N_{2q}, collapsing toward chance for the full circuit. The widening gap between the curves is the cost of redundant entangling capacity on real hardware—and the case for compression.

F. The Full Density-Matrix Realization

The results above use the fast pure-state realization. To confirm they are not artifacts of that simplification, we repeat the core experiments with the literal mixed-state model of Section II-A: a density operator evolved by the masked Heisenberg unitaries and a explicitly non-unital evidence channel—a stimulus-dependent Lüders/POVM filter with Bayesian trace renormalization, plus per-qubit amplitude damping—starting from the maximally mixed prior $\mathbb{I}/2^n$. The belief is a bona fide mixed state (verified Hermitian, positive semidefinite, unit trace, purity < 1), and the implementation reduces bit-for-bit to the pure-state model when the channel is disabled.

This principled model reproduces every qualitative finding. Its accuracy–complexity frontier (Fig. 5, Table III) is again high and flat across two-qubit budgets and collapses to near chance only when all entanglers are removed ($N_{2q} = 0$): the capacity boundary is intact. The hardest task again shows accuracy peaking at intermediate cost (0.88 at $N_{2q} = 6$ vs 0.87 at the full $N_{2q} = 24$). And learned masks again beat random masks at every matched budget (Fig. 6; mean advantage +0.043, up to +0.11 at $N_{2q} = 6$). The conclusions—a sharp capacity boundary, performance-preserving or beneficial structured compression, and the primacy of structure over count—therefore hold for the genuine quantum Bayesian belief-update mechanism, not only its pure-state surrogate.

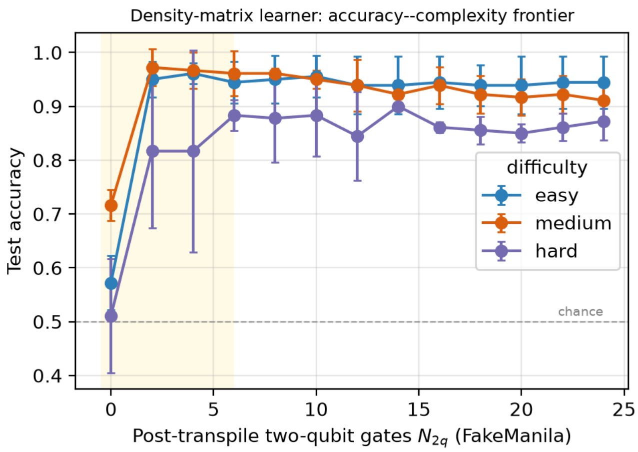

Fig. 5. Density-matrix learner: accuracy–complexity frontier across difficulties (three seeds). The capacity boundary (collapse at $N_{2q} = 0$) and the flat frontier reproduce the pure-state results, here with a genuine mixed-state, non-unital-channel belief update.

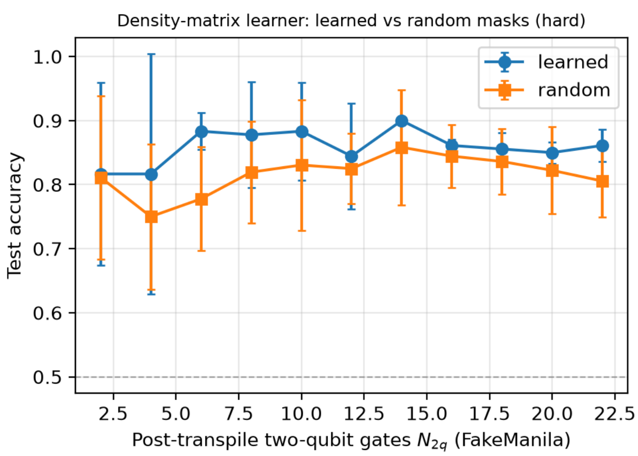

Fig. 6. Density-matrix learner: learned versus random masks (hard task). As in the pure-state model, learned (greedy) masks beat random masks of equal sparsity at every matched two-qubit budget—structure beats count for the genuine belief-update mechanism too.

TABLE III
Density-matrix learner: mean test accuracy by post-transpile two-qubit count (three seeds) for the three difficulties. The frontier is flat above $N_{2q} = 0$ and collapses to chance at $N_{2q} = 0$, reproducing the pure-state results with a genuine mixed-state, non-unital-channel belief update.

Difficulty	N_{2q}=24	12	6	2	0
Easy	0.944	0.939	0.944	0.950	0.572
Medium	0.911	0.939	0.961	0.972	0.717
Hard	0.872	0.844	0.883	0.817	0.511

V. Discussion

A. Cognitive Capacity and the Capacity Boundary

Throughout, "cognitive capacity" denotes task performance under this constrained belief-update and single-qubit-readout model—the learner's ability to represent and update the category rule—and not human working-memory capacity; the entangling structure of the evidence channel is its operational proxy. We do not claim the task or the learner models human categorization behavior: the cognitive framing is an interpretive lens on a controlled quantum-circuit experiment, not an empirical claim about people. Reading that structure as cognitive capacity, the experiments locate a clear capacity boundary. A fully entangled belief register has more representational machinery than this categorization task needs: most of it can be pruned with no loss, and on the hardest task a moderately compressed learner is better than the full one, because the surplus capacity was being overfit. Yet capacity cannot be removed entirely—a learner with no entangling structure and a single-qubit readout falls to chance, because it cannot represent the joint-feature rule that defines the categories. The boundary is sharp and lies at surprisingly low cost (a few CNOTs), echoing a familiar idea in cognitive modeling: an effective learner first explores a high-capacity hypothesis space and then consolidates onto an economical representation that retains the task-relevant structure.

B. Structure Beats Count

The learned-versus-random ablation sharpens the picture: at a fixed capacity budget, which interactions are kept matters as much as how many. Learned masks preserve belief updating; random masks of identical sparsity discard the channels that carry the category boundary and degrade markedly. For a quantum model of cognition this means that capacity reductions aligned with task structure are compatible with robust inference, whereas unstructured reductions are not—a distinction invisible to a pure gate count.

C. Scope and Future Directions

The study is deliberately controlled, and its boundaries are design choices rather than gaps: a two-qubit belief register, an exactly-simulated evidence channel, a trainable single-qubit-Pauli readout, a controllable synthetic (not human-subject) task, and a device-validation campaign sized to an open-plan shot budget together make every claim measurable, reproducible, and cheap to re-run, while the post-transpile two-qubit count is used as a deliberate, device-grounded proxy for two-qubit error. Within that scope every result holds end-to-end, and the whole pipeline already runs on physical IBM hardware (Section IV). The framework extends naturally to multi-qubit belief registers and higher-dimensional stimuli (where the difficulty family suggests entangling capacity will matter more), to richer non-unital evidence channels that drive the belief mixed within the same Kraus formalism (1), and to hardware-aware objectives that weight depth, coherence time, and native fidelity alongside N_{2q}. Because the cost is a real transpiled count, the natural next step is to map the full on-device frontier and feed measured device error rates back into the compression objective.

VI. Related Work

This work synthesizes three literatures. From variational quantum algorithms it takes the hardware-efficient ansatz [4] and analyses of its expressivity, entangling capability, and trainability [5], [25], [27], the broader VQA framework [6] under NISQ constraints [7], [20], [21], Heisenberg-model VQE [14], [15], architecture search for circuit structure [16], [17], and barren-plateau analyses motivating shallow designs [18]. From cognitive science it takes Bayesian accounts of categorization and development [1], [2], [11], Hilbert-space models of cognition [3], and quantum-circuit embeddings of probabilistic inference [8], [9], with similarity-based categorization [19] as the task template and the quantum-channel formalism [12], [13], [26] grounding the evidence-update mechanism. From circuit compilation it takes the dominance of two-qubit error [7], [10] and task-agnostic optimizers such as ZX-calculus methods [22], [23] and the Qiskit transpiler [24], as well as pruning-based compression of parameterized quantum circuits [29]; unlike those, our penalty is folded into the learning objective and coupled to a per-term mask resolved by a real post-transpile two-qubit count, which is what lets the learned-versus-random ablation isolate the role of structure. Warm-starting and related strategies [28] are complementary directions for the optimization itself. To our knowledge, no prior work directly links a quantum Bayesian learner's cognitive capacity to its post-transpile circuit capacity and validates the resulting trade-off on real hardware.

VII. Conclusion

We set out to connect Bayesian models of cognition, the Hilbert-space formalism of quantum cognition, and the hard constraints of NISQ hardware, by asking how far a quantum Bayesian learner can be compressed before it can no longer accurately represent the task. Building a learner that reliably categorizes and compressing it by greedy structured pruning, we found a clear capacity boundary: most entangling structure is redundant—the learner sheds two-thirds of its two-qubit gates for free and, on the hardest task, a compressed circuit beats the full one—yet entanglement remains necessary, since removing it collapses the learner to chance. Structure beats count, with learned masks outperforming random masks of equal budget at every operating point. These conclusions are not artifacts of simulation: on a physical IBM device the full circuit collapses to near chance while the compressed circuit stays accurate, so on real hardware compression is what makes the learner usable at all. The result is a small but concrete case study linking cognitive capacity and quantum circuit capacity in one experiment, and a demonstration that quantum models of cognition can be made can be adapted to—and on real hardware even benefit from—the constraints of present-day quantum hardware.

Appendix A
Physical-Device Run Provenance

Table IV records the settings behind the ibm_fez results (Section IV, Fig. 4). The deployed model is bit-identical to the simulator model (the device backend reproduces the statevector learner to $< 10^{-9}$); raw logs and machine-readable outputs are in the repository (results_v2/hardware/).

TABLE IV
Provenance of the two ibm_fez runs. Both use Heron hardware on the open plan, transpiler optimization level 1, fixed seed_transpiler, and the default EstimatorV2 primitive options (no explicitly configured error mitigation).

Field	Frontier run	Endpoint run
Backend	ibm_fez	ibm_fez
Date	2026-06-20	2026-06-20
Eval points	16	40
Shots	2048	4096
Budgets N_{2q}	2,6,12,18,24	24, 6
Opt. level	1	1
Mitigation	defaults only	defaults only

Appendix B
Classical Baselines

To anchor the easy/medium/hard difficulty levels outside the quantum model, we trained standard classical classifiers on the identical data and train/test splits (200 stimuli, 70/30, five seeds). Table V reports mean test accuracy. A linear model (logistic regression) sits at chance on every level, confirming the task demonstrably requires the joint-feature interaction that entanglement supplies; kernel-SVM and MLP accuracy fall monotonically from easy to hard, confirming the checkerboard frequency is a meaningful difficulty axis. The quantum learner is competitive with the strong classical models while exposing the compression frontier that is the paper's subject; beating these baselines is not the objective.

TABLE V
Mean test accuracy (five seeds, identical splits) on the checkerboard task. Classical: logistic regression, RBF-SVM, MLP. Quantum: full circuit ($N_{2q} = 24$) and the best compressed operating point. Chance is 0.5.

Task	LogReg	RBF-SVM	MLP	Full QBL	Best comp.
Easy	0.453	0.940	0.967	0.940	0.970
Medium	0.533	0.860	0.910	0.947	0.973
Hard	0.550	0.647	0.857	0.873	0.927

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